

| Question |     | Answer          | Mark | Mark scheme                                                                                                                                          | Additional guidance                                                                                                                           |
|----------|-----|-----------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 15       | (a) | $5n^2 + 2n - 4$ | M1   | for a correct start to a method to find the $n$ th term, eg constant 2nd differences and $n^2$<br><b>OR</b> states $2a = 10$ <b>or</b> $3a + b = 17$ | Need to see constant 2nd difference found and $n^2$<br>Condone use of a different variable throughout<br>$a = 5$ <b>or</b> $b = 2$ implies M1 |
|          |     |                 | M1   | for working with $5n^2$ , eg $5n^2$ , and sequence $-2, 0, 2, \dots$<br><b>OR</b> states $2a = 10$ <b>and</b> $3a + b = 17$                          | $5n^2 + 2n$ implies M2<br>$5n^2$ is implied by $5, 20, 45, \dots$<br>$a = 5$ <b>and</b> $b = 2$ implies M2                                    |
|          |     |                 | A1   | for $5n^2 + 2n - 4$ oe                                                                                                                               | Condone $+ - 4$                                                                                                                               |
|          | (b) | 1.2             | M1   | for substituting values, eg $4 = k \times 9 + k$ <b>or</b> $4 = k(9 + 1)$ <b>or</b> $(k = 0.4)$                                                      |                                                                                                                                               |
|          |     |                 | M1   | for $(u_3 =)$ “0.4” $\times 4 +$ “0.4” $(= 2)$                                                                                                       |                                                                                                                                               |
|          |     |                 | A1   | for 1.2 oe                                                                                                                                           |                                                                                                                                               |